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A LAGRANGIAN RELAXATION APPROACH TO THE CLASSROOM ASSIGNMENT PROBLEM*

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ABSTRACT

This paper describes one of the algorithms in a new timetabling and student registration system that was implemented at the University of Waterloo (Ontario, Canada) in May, 1987. We are given a set of class meetings that has already been assigned to specific days and time periods of the week, and a collection of available classrooms on campus. We must determine an acceptable assignment of classes to rooms based on a variety of factors that measure the desirability of a particular assignment.

The problem is subdivided into two separate components. Given a function c_{ij} that represents the cost of assigning class i to room j , we present a heuristic algorithm based on a Lagrangian Relaxation for solving the multiple period assignment problem [MPAP]. We then describe a cost function model that will be shown to reasonably reflect the general preference structure of the faculty and the administration. Computational results on 1400 courses at the University of Waterloo are discussed.

RÉSUMÉ

Ce document décrit un algorithme qui fait partie d'un logiciel pour l'établissement des horaires et l'inscription des étudiants utilisé depuis mai, 1987 à l'Université de Waterloo (Ontario, Canada). On suppose qu'un ensemble de classes a déjà été affecté à des jours et à des heures particulières et qu'un certain nombre de salles sont disponibles sur le campus. Nous désirons déterminer une affectation de classes aux salles selon un groupe de facteurs exprimant l'avantage de ladite affectation.

Le problème se divise en deux parties. En premier lieu, nous proposons un algorithme heuristique basé sur la relaxation lagrangienne pour résoudre le problème d'affectation à périodes multiples, étant donné la fonction c_{ij} du coût de l'affectation de la classe i à la salle j . Nous décrivons ensuite un modèle de coût représentant assez bien la préférence générale des professeurs et de l'administration. Enfin, des calculs sont fournis pour un problème de 1400 cours à l'Université de Waterloo.

1. INTRODUCTION

The University of Waterloo has recently completed the staged implementation of a complete student timetabling and registration system. The first major phase of the system consists of a computerized timetabling subsystem that is used to determine the times at which each class should be held.

After the times have been established, the individual classes must be assigned to the available rooms on campus. The timetabling system ensures that there are enough rooms of appropriate size to accommodate all of the classes at all time periods. However, timetabling does not consider the special characteristics of individual rooms (*e.g.* location, seating type, special facilities, *etc.*). The Classroom Assignment algorithm tries to find an acceptable room for each class at the specified time periods.

There are a variety of characteristics that define a good assignment of classes to rooms. A cost function, c_{ij} describes the cost of assigning class i to room j . A cost of zero represents a perfect match and higher costs measure the relative undesirability of the given assignment. Any unacceptable assignment is given an infinite cost. (*e.g.*:

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when the room is too small for the class.) The problem of determining an appropriate cost function is crucial to the success of the algorithm. The cost model that has been implemented at Waterloo is described in Section 4. For the moment, it will be assumed that these costs have been calculated for all classes and rooms.

Observe that, if all classes are the same length and there is no overlap of times, then the problem is trivial. For example, if all classes are one hour in length and always begin on the hour, then each hour of the day can be solved as a separate assignment problem. The solution is always feasible and optimal (assuming that a feasible solution exists).

In the Waterloo problem, classes can be 1, 1.5, 2, 3 or 4 hours in length beginning every half hour. The majority of lecture style classrooms on campus are controlled by the central Scheduling Office. Most laboratories and special purpose space are controlled separately by the individual departments. The room assignment algorithm is concerned with the central pool of classrooms and instructors must specify which of their meets require Scheduling Office space. This has the effect of removing many of the three and four hour laboratory classes from the problem. However, there are still a large number of classes that are longer than one hour in length.

2. MATHEMATICAL FORMULATION OF THE PROBLEM

The usual formulation of the problem is expressed in terms of the variable x_{ij} , a 0-1 valued variable that is equal to 1 when class i is assigned to room j . The problem can be described in the form of a linear programming problem.

$$\min_{x_{ij} \in \{0,1\}} \sum_{i=1}^n \sum_{j=1}^m c_{ij} x_{ij}$$

subject to:

$$\begin{aligned} \sum_{j=1}^m x_{ij} &= 1 && \text{for } i = 1, 2, \dots, n \\ \sum_{i \in P_k} x_{ij} &\leq 1 && \text{for } j = 1, 2, \dots, m \text{ for all periods } k \end{aligned}$$

where P_k represents the set of all classes that meet in period k . The first constraint ensures that every class is assigned to some room while the second constraint prevents any room from having more than one class in the same period. Carter & Tovey (1989) have shown that this problem is NP-Hard even when the problem has only two periods.

Several heuristics for solving this problem have appeared in the literature in recent years. Mulvey (1982) combines the problems of assigning classes to times and to rooms using an underlying partial network formulation. However, his approach is limited to very small problems (around ten classes) and doesn't appear feasible for large practical problems.

Ferland & Roy (1985) also suggest solving both problems, but they solve them in sequence and iterate if necessary. Both subproblems are modelled as quadratic assignment problems where a conflict between two assignments is "penalized" in the objective function rather than constrained as in the usual linear programming formulation. They search for all possible single and double interchanges of classes to find a locally optimal solution. They present computational results using actual data from the Université de Montréal. The classroom assignment phase will be relatively fast, but it is difficult to analyze the quality of the final solution.

Glassey & Mizrach (1986) describe a heuristic that was implemented at the University of California at Berkeley in 1983. Their method employs a greedy "list-processing" strategy. They choose the half-hour period of the week with the highest ratio of unscheduled classes to vacant rooms. They then assign all classes in that period beginning with the largest ones first. When a class is assigned to a room in one period, it is also "booked" in that room for all other required periods. Within each period, they try exchanges of up to three classes simultaneously, but they don't alter rooms fixed in previous periods. As with Ferland & Roy, the approach is relatively fast, but difficult to assess. In particular, the method cannot guarantee that a feasible solution will be found when one exists.

Gosselin & Truchon (1986) solve each day separately and model the problem as a linear program. In order to keep the number of variables and constraints within reasonable limits, they aggregate the rooms into groups of similar size and facilities and aggregate classroom requests by size, facilities and time period. They tested their procedure using actual data from Université Laval in Québec.

They discovered that the corresponding linear program always produced integer solutions. This result can be attributed to their use of a cost function based primarily on the size of each room. When the classroom assignment problem is restricted to such monotonic cost functions, it remains NP-Hard, but probably requires considerably more than two or three periods (see Carter & Tovey (1989)). Since the cost function proposed at Waterloo does not have this property, the LP solution will not, in general, be integer valued. Moreover, the size of the corresponding LP is prohibitive.

For the Waterloo problem, in order to reduce the overall problem size, each day of the week was solved separately. In addition, we imposed a division of each day into three parts: morning, afternoon and evening. We divided the day at 12:30 p.m. and again at 6:30 p.m. and insisted that no class run over these breakpoints. This was not considered to be a serious restriction at Waterloo for lecture classes, and allowed us to decompose the original problem into 14 subproblems: morning, afternoon and evening each day with no Friday evening classes. (For classes that meet several times a week, each meeting is treated as a separate class. However, we modify the cost of subsequent assignments to encourage the use of the initial room).

The present approach can be considered as a variation of Mulvey's (1982) idea in that we find a subproblem that has a network structure and then penalize the network at each iteration to account for infeasibilities. In particular, the problem is formulated as a series of assignment problems with a set of side constraints. Each period of the day (e.g. each half hour time period) can be considered as a separate two-dimensional assignment problem. If the problem was solved in this fashion, it is entirely possible that classes would be required to get up and move to a new room at the end of each period. Clearly, this is an unacceptable situation. Therefore, the problem must include a set of "Room Jumping Elimination" constraints that will force classes to stay in the same room in each period of the day.

These additional constraints can be expressed in a variety of ways. The simplest formulation can be stated as follows: If class i is assigned to room j_1 in time period k_1 and to room j_2 in time period k_2 where $j_1 \neq j_2$ and $k_1 \neq k_2$ then the assignment is infeasible. Therefore, for each class i and for each infeasible combination of rooms at different times, a "Room Jumping Elimination" constraint can be included.

Define x_{ij}^k to be equal to 1 if class i is assigned to room j in period k . The required constraints can then be defined as:

$$x_{ij_1}^{k_1} + x_{ij_2}^{k_2} \leq 1 \quad i = 1, 2, \dots, n \quad \text{for all } j_1 \neq j_2 \quad \text{for all } k_1 \neq k_2$$

where n represents the number of classes in a subproblem. The number of constraints can be reduced by considering only adjacent values of k_1 and $k_2 = k_1 + 1$. However, for typical values in the Waterloo subproblems with over 200 classes, with 25 feasible rooms and 4 periods for each class, this formulation would still require approximately 180,000 of these constraints. Clearly, explicit enumeration of all possible constraints is not a viable approach.

3. THE ALGORITHM

3.1 The "Lagrangian" Phase

The procedure that was adopted uses a Lagrangian Relaxation approach with constraint generation. The problem is first solved as a sequence of independent assignment problems, one for each period. These solutions are then tested for feasibility, and if any classes are assigned to different rooms in consecutive periods, the corresponding "Room Jumping Elimination" constraint is constructed and appended to the objective function in the form of a penalty term. Define:

$$f(x) = \min_{x_{ij}^k \in \{0,1\}} \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^p c_{ij} x_{ij}^k$$

where n is the number of classes, m is the number of rooms and p is the number of periods. This multiple period assignment problem [MPAP] has two sets of constraints.

$$\begin{aligned} \sum_{j=1}^m x_{ij}^k &= 1 && \text{for all periods } k \text{ and } i \in P_k \\ \sum_{i=1}^n x_{ij}^k &\leq 1 && \text{for all } j, k \\ x_{ij}^k &= 0 && \text{for all } i \notin P_k \end{aligned}$$

The set P_k is defined as the set of all classes that require a room in period k . The first constraint states that each class must be assigned a room in each required period and the second constraint restricts each room to at most one class in each period.

Denote the collection of "Room Jumping Elimination" constraints by $Ax \leq \bar{1}$ where $\bar{1}$ represents a vector of 1's. Then, the Lagrangian function can be written as:

$$L(x, u) = \max_{u \geq 0} \min_{x_{ij}^k \in \{0,1\}} f(x) + u^T \{Ax - \bar{1}\}$$

subject to:

$$\begin{aligned} \sum_{j=1}^m x_{ij}^k &= 1 && \text{for all periods } k \text{ and } i \in P_k \\ \sum_{i=1}^n x_{ij}^k &\leq 1 && \text{for all } j, k \\ x_{ij}^k &= 0 && \text{for all } i \notin P_k \end{aligned}$$

This formulation can be interpreted as a penalty function for violating the room jumping constraints. Each element of the vector u is called a *Lagrange multiplier* and represents a penalty for violating the corresponding constraint. A violated constraint results in a positive value of $(ax - 1)$, which induces a positive penalty. It is well known that:

$$f(\hat{x}) \leq L(\bar{x}, \bar{u}) \leq f(x^*)$$

where $\hat{x}$ is the optimal solution to the multiple period assignment problem, $L(\bar{x}, \bar{u})$ is the optimal solution of the Lagrangian and x^* is the required optimal solution to the original classroom assignment problem. (Refer to Fisher (1981) for an excellent survey and tutorial on Lagrangian Relaxation Methods and a general proof of the preceding inequality.)

Observe that, if we choose any fixed non-negative value for u , the problem can be decomposed into p subproblems: one for each value of k . Each subproblem is a simple assignment problem. The algorithm for solving the Lagrangian employs this property iteratively. Begin with $u = 0$ and solve the p assignment problems. Find the violated "Room Jumping Elimination" constraints, increase the corresponding penalties, and resolve the p assignment problems.

The routine can be improved with the use of "Generalized Room Jumping Constraints". In the Waterloo problem, we observed that the single period assignment problems generally exhibit a large number of alternate optimal solutions.

Consider a particular course i that must be assigned in periods 1 and 2. Suppose that course i is assigned to room j_1 in period 1, and room j_2 in period 2. We can determine all possible alternate rooms for class i in period 1 by adding a sufficiently large value to the current cost of assigning class i to room j_1 and re-solving the period 1 assignment problem. If the cost of the new assignment solution is the same as the previous cost, then we have an equal cost alternate room. This element also can be penalized and the process repeated until the cost of the assignment solution (for period 1) increases. Define Pen_1 to be the amount of this increase. Pen_1 is interpreted as the minimum increase in cost that must be incurred if we don't use any of the alternate optimal rooms.

Repeat the same process for period 2 and define Pen_2 to be the corresponding cost for not using any of the second period alternate rooms. Let R_1^i and R_2^i denote the sets of alternate optimal rooms for class i in periods 1 and 2 respectively.

We now compare the two lists of alternate rooms. Observe that every feasible solution to the original problem must use the same room in both periods. Clearly, if the two lists have no rooms in common, then any optimal solution must use at least one room that is not in the list of alternates in at least one period. The cost of using a room which is not in the set $R_1^i (R_2^i)$ is at least $Pen_1 (Pen_2)$. We say that a necessary condition for the multiple period assignment problem to contain a solution to the original classroom assignment problem is that every pair of alternate optimal room sets for each class i must have at least one room in common.

Suppose some pair have no rooms in common. The cost of any feasible solution must incur an increase of at least:

$$Pen = \min\{Pen_1, Pen_2\}$$

This feasibility condition can be described by the constraint:

$$[\text{TYPE I}] \quad \sum_{j \in R_1^i} x_{ij}^1 + \sum_{j \in R_2^i} x_{ij}^2 \leq 1$$

Using this constraint, we can construct the corresponding Lagrangian function as follows:

$$L(x, u) = \min_{x_{ij} \in \{0,1\}} \left[f(x) + u \left\{ \sum_{j \in R_1^i} x_{ij}^1 + \sum_{j \in R_2^i} x_{ij}^2 - 1 \right\} \right]$$

for some fixed scalar multiplier $u \geq 0$, subject to the original multiple period assignment constraints.

The behaviour of $L(x, u)$ as a function of u is highly predictable. Observe that, for a fixed value of u , the function $L(x, u)$ is equivalent to a new multiple period assignment problem with the c_{ij} terms increased by the scalar u in period 1 (period 2) for all $j \in R_1^i$ ($j \in R_2^i$) and then the constant u is subtracted from the optimal value.

Without loss of generality, assume that $Pen.1 \leq Pen.2$. For $0 \leq u \leq Pen.1$, the available set of alternate optimal solutions will remain the same in both subproblems. The cost of the optimal assignment in both periods will increase by u and, after subtracting the constant u , the solution of $L(x, u)$ will increase by precisely u .

When $u > Pen.1$, we know that there is at least one new alternate room in period 1 that will cost only $Pen.1$ more than the original solution. Hence, the penalized rooms in the set R_1^i will no longer be alternate candidates. Increasing u beyond $Pen.1$ will have no further affect on the cost of the period 1 solution. Similarly, increasing u beyond $Pen.2$ will force the period 2 subproblem to choose some room j not in the set R_2^i and that component of the cost will stop increasing. To summarize:

$$L(x, u) = \begin{cases} f(\hat{x}) + u & u \leq Pen.1 \\ f(\hat{x}) + Pen.1 & Pen.1 \leq u \leq Pen.2 \\ f(\hat{x}) + Pen.1 - (u - Pen.2) & Pen.2 \leq u \end{cases}$$

The increase in $L(x, u)$ is maximized by choosing $Pen.1 \leq u \leq Pen.2$. In particular, we will select $u = Pen.1$. At this point, the penalized assignment problem in period 1 has at least one additional alternate optimal room (by definition of $Pen.1$). The set R_1^i can be increased as before to include all such alternate rooms, and the necessary condition for feasibility can be tested again to see if any of the new rooms correspond to alternate assignments in period 2. If not, a new constraint can be generated, and the Lagrangian function can be augmented with a second penalty term. Note that, if $Pen.1 = Pen.2$, then period 2 will also have additional alternate rooms.

This procedure is finite since each iteration adds at least one new room to one of the two periods. Furthermore, the values of the penalty terms, u , are always integer valued. This is important for the assignment algorithm and for the convergence of the method. Since $L(x, u)$ increases by an integer value in each iteration, and since the Lagrangian is bounded above by $f(x^*)$, the algorithm must be finite.

In computational results on practical problems, it was observed that this process normally required only one iteration to generate a feasible room overlap. However, for a few courses, several iterations were required. For these classes, we noticed that $Pen.1$ was much smaller than $Pen.2$. For example, consider the situation illustrated in Figure 1.

	Room:									
	1	2	3	4	5	6	7	8	9	10
Period 1	X	10	11	12	13	14	15	16	17	18
Period 2	0	10	11	12	13	14	15	16	17	18

Figure 1. An example of slow convergence

In period 1, room 1 is unavailable and room 2 is optimal. Pen_1 is 1, the added cost of using room 3. In period 2, room 1 is optimal, and Pen_2 is 10, the added cost of using room 2. The Lagrangian routine as described above will consecutively add rooms 3, 4, 5, ..., 10, one at a time, to the list of period 1 alternates before room 2 is added to the solution of period 2, the obvious feasible solution.

In order to compensate for this observed behaviour, a second type of Lagrangian constraint was incorporated whenever Pen_2 is sufficiently larger than Pen_1 . The modified routine has the desirable property that it either produces an increase in $L(x, u)$ of Pen_2 in one iteration, or it guarantees an overlapping room assignment in two iterations of the Lagrangian. In this case, we introduce the following constraint:

$$[\text{TYPE II}] \quad \sum_{j \in R_2^i} x_{ij}^2 + \sum_{j \notin R_2^i} x_{ij}^1 \leq 1$$

This constraint, when included in the Lagrangian function, has the effect of penalizing every room for course i in either period 1 or period 2.

Again, the value of the Lagrangian is very predictable. By choosing a trial penalty value of $u = Pen_2$, the cost of the period 2 assignment increases by precisely Pen_2 , and at least one new room is added as an alternate solution. For the period 1 problem, the cost of all rooms *except* $j \in R_2^i$, is increased by Pen_2 . Therefore, when the period 1 problem is re-solved, either the optimal room $j^* \in R_2^i$, or the cost of the assignment must increase by Pen_2 . In the latter case, $L(x, u)$ has increased by Pen_2 , and we can continue checking for overlap. In the former case, let Pen_{j^*} be the increase in the cost of the period 1 assignment. Clearly:

$$Pen_1 \leq Pen_{j^*} \leq Pen_2.$$

Redefine $u = Pen_{j^*}$ and repeat the above Lagrangian solution. It is easily verified that the new $L(x, u)$ increased by precisely Pen_{j^*} and that the new solutions overlap in the set of rooms $j \in R_2^i$.

Constraints are added to the Lagrangian function one at a time to preserve integrality and predictability. It is easy to show that using two or more constraints in the same iteration (with one step size u) can lead to fractional values of u , and hence, fractional assignment entries c_{ij} in the penalized subproblems. Moreover, using multiple constraints can actually result in smaller increases in the Lagrangian function. The constraints with a low value of Pen_2 unnecessarily restrict the size of the improving step.

The preceding Lagrangian phase produces a transformed room assignment problem with the following properties:

1. Every pair of time periods for each class has at least one alternate optimal room in common.
2. The function value of the optimal solution to the room assignment problem is bounded below by the function value of the corresponding penalized Lagrangian.

This Lagrangian function has the so-called *integrality property*. Solving the Lagrangian without the integer restriction would still produce an integer solution. Therefore, the value of the optimal Lagrangian solution will always be equal to the value of solving the multiple period assignment problem as a linear program with the addition of the "Room Jumping Elimination" constraints. (Refer to Fisher (1981) for details.) Hence, we expect that there will be a *gap* between the Lagrangian function value and the optimal room assignment solution, and we will evaluate this gap in our computational results.

3.2 The "Solution" Phase

For small practical subproblems (*i.e.*, with one or two periods only), the Lagrangian phase often produces a feasible solution to the room assignment problem. For the Waterloo problem, these were always optimal solutions to the original MPAP (although this will not be true in general). For larger problems, it is necessary to search for a feasible answer.

The Solution phase employs a straight-forward heuristic using the penalized Lagrangian cost matrices above. For each course, we construct a list of rooms which are acceptable alternates in all periods. Observe that the Lagrangian phase guaranteed that every pair of periods has at least one common acceptable room, but some classes may not have any acceptable rooms in all periods.

The algorithm assigns rooms in order of the least number of remaining acceptable alternates. When a room is assigned, it is removed from the list of acceptable alternates for all other classes which have an overlapping time period.

This procedure continues until some class has no acceptable alternates. We then perform a backtracking routine that searches for a feasible reassignment of the classes which are currently assigned to the original acceptable rooms for the current class. The algorithm enumerates all possible alternating assignments using a tree data structure until it finds a class that can be reassigned to a new acceptable alternate. The resulting path from this class to the root of the tree defines a sequence of reassignments that will create an acceptable alternate for the original course.

If no feasible reassignment is found, we then increase the number of acceptable alternates for all classes which were included in the tree. This is accomplished by redefining an acceptable alternate to include rooms with the "Next Lowest" cost above the original acceptable cost. Having increased the number of acceptable alternates for these classes, the solution procedure is repeated from the beginning with all previous assignments removed.

In the current implementation of the program, we compute and store the next lowest cost rooms for all classes and append these to the original list of acceptable alternates with their associated cost. The solution procedure still tries to assign all classes to the acceptable alternate rooms. However, in the backtracking routine, if no acceptable reassignment is found, the next lowest cost reassignment is used. In this way, we found that it was never necessary to perform a second solution iteration. The backtracking procedure was always able to locate a reasonably inexpensive alternate assignment.

4. THE COST OF AN ASSIGNMENT

The preceding algorithm assumed that we had a cost c_{ij} for assigning each class i to each room j . Most of the papers in the literature assume that cost is a fairly simple function of space (empty seats). In practice, this criterion is not the most significant factor. Glassey & Mizrach (1986) implemented a more realistic function that takes distance, facilities and space into consideration. In our approach, we have extended this to try to capture the preferences of the manual room scheduler.

Assigning a specific class to a particular classroom can be described in terms of a multiple objective decision making problem. There are a number of criteria that can be used to define a good assignment. The cost of a given assignment is based on a combination of these factors.

In order to simplify the model, it was assumed that the cost components are additive. Total cost is determined by computing a cost for each individual factor and then adding them. This is equivalent to assuming that the "decision maker's" underlying utility function is linear. However, most of the cost factors in the model are described

using only a small number of discrete values. (e.g., most of the rooms assigned will have distance cost 0 or 10). Consequently, we felt that the linear utility assumption did not represent a serious shortcoming of the model.

For the purpose of evaluating tradeoffs, the Scheduling Office was considered as the "Decision Maker". First, we determined a reasonable distance/building preference cost. All other costs were developed by presenting the Scheduling Office with simple scenarios in terms of one or two classes and asking them to choose. In this way, we could evaluate the importance of distance preference compared to all other criteria.

Throughout the design of the system, one major criterion was to limit the necessary input that would be required from the individual departments. In particular, the factors that define an acceptable assignment will vary from one instructor to the next. Therefore, most of the cost factors are based upon some average or standard concept of suitability. For example, distance/building preference is established for each subject group. All "Math" courses are assumed to have the same preference. When necessary, these defaults can be overridden on an individual course basis.

4.2 Distance/Building Preference

In an early version of the model, each subject was assigned a preferred location on campus based on the location of the home department. The distance between this preferred location and the actual location of the classroom was calculated and squared as a measure of cost. This measure is the same as the one used by Glassey & Mizrach (1986).

Although this appeared to be reasonable, it did not adequately reflect the preference structure that exists in an academic environment. In particular, in discussions with faculty, it was discovered that distance is not a symmetric function. Faculty teaching subject *A* do not like to travel to building *B*, but faculty in the corresponding subject *B* are quite happy to teach in building *A*. This can occur for a variety of reasons. Perhaps the facilities in building *A* are considered better, or the seating is more appropriate to certain purposes. In at least one case at Waterloo, a department had three non-adjacent preferred buildings since the predominant seating type in those buildings (tables and chairs) was considered desirable for their purposes.

The current model asks each department to define up to four groups of buildings in order of preference. Group 0 includes all of the preferred buildings for the subject. Group 1 contains a list of buildings that can be used if there are no appropriate rooms available in group 0. Groups 2 and 3 are defined analogously. Any building that is not listed in this table must never be used for that subject. It is assumed that people would rather change the time of their course than travel to an unlisted building.

Some specific courses within a subject group will have different building preferences. For each such course, a unique preferred building (or group of buildings) can be defined and stored permanently on the course record file. For these courses, the preferred building(s) becomes group 0; group 0 from the table becomes group 1, etc.

To convert this to a numeric cost, the group number is multiplied by 10. This produces costs ranging from 0 to 40. All of the remaining cost factors are defined relative to this preferred building cost. *i.e.*, How far would people be willing to travel in order to have a desired resource?

4.3 Space Utilization

The primary consideration for space utilization is that every classroom should have enough seating stations to accommodate all members of the class. Although this appears obvious, it presents some tricky problems. At Waterloo, rooms are assigned up to three months before classes begin and accurate estimates of class size are not available.

Each course at Waterloo has an associated Section Limit. The system will restrict enrolment in that course (or course section) to that value. It seemed reasonable to assume that every classroom should be large enough to accommodate the section limit. Unfortunately, section limit is an absolute maximum number of students that can be allowed in the course. It is often chosen for academic reasons such as available resources (lab equipment, teaching assistant time, instructor's capacity, *etc.*) Section limit has little resemblance to actual enrolment. At Waterloo, we discovered that 80% of all courses were using rooms that were *smaller* than the section limit!

Various schemes for estimating enrolment based on section limit, historical enrolment and observed variances were rejected. It was felt that individual deviations were too unpredictable to be useful. The assignments would either waste a lot of space, or they would produce an unacceptable number of inappropriate matches.

The present model asks people to specify a 'Minimum Room Size Required'. This information is carried forward from year to year and changed on an exception basis. If not specified, the section limit is used as a default. The system will not assign any room that doesn't contain enough seats to accommodate 'Room Size Required'.

The second aspect of space utilization is that the assigned room should not contain a large number of unused seats. This is not a serious consideration after the school term is well under way, but the computerized room assignment is typically followed by a period of adjustment. People will change rooms, change times and, as final enrolment figures develop, some classes must move to larger facilities. If we fill up all of the large classrooms with small classes, we can unduly restrict the flexibility of the adjustment phase.

In the current version of the model, a room assignment is given a penalty of 1 for every 10 empty seats. In addition, a cost of 5 is added if the class would use less than 60% of the seats, and a cost of 10 for using below 30% of the space. The intention is that a large class using less than 60% of the available seats may be forced to move to a less preferred building to get an appropriate sized room. Similarly, *any* class using less than 30% of the available space could move to a less preferred building if a proper sized room was available. In the computational results section, it will be shown that these rules did not have a significant impact on preferred building assignments.

4.4 Special Facilities

Individual instructors may optionally request one or more special room facilities. Many of these facilities such as projection facilities, television and computer terminals, are absolute requirements. Any room that does not satisfy these requests will not be considered for assignment. Some special facilities can be substituted at a cost. These facilities are considered desirable rather than required. For example, at Waterloo, some classrooms have seminar (conference) style seating. Although this arrangement is considered useful in certain courses, it was felt that people would not be willing to walk across the campus for seminar seating if there was a room with tables and chairs available in their preferred building. Consequently, a penalty of 20 was applied to rooms with table and chair seating when the class requested seminar style seating.

Similar substitutions occur when people request a large blackboard. A medium sized blackboard can be substituted at a penalty, but a room with a small blackboard is considered unacceptable.

4.5 Preferred Rooms

Instructors are also allowed to specify a preferred room (or group of rooms) for their class. These rooms will be automatically given a cost of zero in the assignment algorithm. All other non-preferred rooms will have their total cost increased (penalized) by five.

This has the effect of giving a high preference to the indicated rooms. However, the penalty is not high enough to force another course to move to a less desirable building. Again, in the next section, it will be shown that the majority of these requests at Waterloo were satisfied where possible.

4.6 Fragmentation

Recall that the room assignment algorithm solves each half day problem separately. When a course is offered for one hour, three times a week, (Monday, Wednesday, Friday), the algorithm will find the best possible assignment on each of the three days. In the design stage of the routine, this was considered to be reasonable. If the Monday assignment is not in the preferred building, most people would like to have a chance to get back their own building on Wednesday.

When the results of the early tests were shown to faculty representatives, they decided that the resulting level of what they described as room 'fragmentation' was unacceptable. Under the old manual system, the Scheduling Office had always attempted to keep classes in the same room every day. The computer solution was too much of a departure from the norm.

The problem was effectively solved by modifying the costs for the Wednesday problem. If a class received a specific room on Monday, then the cost of any other room on Wednesday is increased by 30. The effect of the penalty is that the assignments will be the same (whenever possible) unless there is a much better room available on Wednesday. In particular, if the Monday assignment is across campus, any rooms in the preferred building will be considered on Wednesday. The same rule applies on Friday. If a class is assigned to two different rooms on Monday and Wednesday, then both of these rooms will be considered as preferred in the Friday problem. Note that, although the Monday, Wednesday, Friday example has been used here, this rule can be applied to any situation involving multiple meet times.

4.7 Non-Standard Timeslots

At the University of Waterloo, the majority of classes use one of a list of "standard timeslots". These groups of periods have been defined such that the different classes will fit together nicely over the week. The use of standard timeslots simplifies the room assignment problem. For example, the standard timeslot "10:30 MWF" fits together with the slot "10:30 TR" (Tues., Thurs.). These classes can share the same room. Many instructors prefer to use non-standard times. However, it is difficult to fit a class that uses "10:30 M, 2:30T" with other classes, although this combination may be appropriate for other reasons.

In the manual room assignment procedure, the Scheduling Office generally followed the rule that classes which used standard timeslots were assigned first. Non-standard classes would get leftover space, possibly in different rooms for each meeting. They felt that this enabled them to find better overall solutions.

In the computerized system, we are not assigning classes in any particular order. The assignment solution effectively assigns all classes at once. However, we felt that we should penalize the use of non-standard timeslots slightly. In particular, if we are faced with a situation where one of two classes must have their assignments fragmented (multiple meets in different rooms), the standard timeslot class should have an advantage, when all other considerations are equal.

To accomplish this, we *reduce* the cost of all room assignments for all non-standard classes by 10%. To appreciate the effect of this reduction, consider two classes that both have the same preferred room and the same alternate room at the same cost. Suppose class 1 is standard and class 2 is non-standard. The cost of assigning both classes to the

preferred room is zero. The cost of relocating the non-standard class to the alternate room is 10% less than the cost of relocating the standard timeslot class. Therefore, in the case of a tie, the non-standard class will move to the less desirable space.

5. COMPUTATIONAL RESULTS

The Classroom Assignment algorithm was implemented at the University of Waterloo in PL/I on an IBM 4341 computer. The algorithm was run in parallel with the manual assignment procedure for the Winter term (Jan.-Apr.) of 1987. The problem involved approximately 1400 courses which included 2192 class meetings in 125 rooms.

There are really two issues that must be addressed in analyzing the results. The first part concerns the effectiveness of the algorithm in solving the mathematical classroom assignment problem. Given any set of costs, we want to estimate how well the algorithm solves the Classroom Assignment problem. The second question deals with the quality of the final solution. In this context, we are more concerned with validating the specific cost function model that was implemented.

5.1 *The Effectiveness of the Algorithm*

The four evening problems were solved easily by the Lagrangian routine. The ten day-time problems contained between 164 and 287 classes in the 125 rooms. Each of these required ten to fifteen iterations of the Lagrangian and one iteration of the Solution phase.

The complete problem was solved in 15 minutes of CPU time. The largest single component of this time was required for solving the individual single period assignment problems. Each problem had an average of 90 rows and 125 columns and required approximately 2 seconds of CPU time. Recall that, each time we add one new alternate room to the list for one class, we must solve the corresponding assignment algorithm. For this problem, we used a special routine that takes advantage of the sparse nature of the cost matrix and included a "RESOLVE" option for finding alternates. The "RESOLVE" routine required 0.1 seconds of CPU time per call. Much of the effort in the design phase of the algorithm concentrated on developing an efficient assignment subroutine and minimizing the number of calls required.

Table 1 summarizes the results of the parallel test in terms of total cost. For the 14 problems, CLASSES is the total number of classes in the subproblem, ASSIGN is the cost of the initial single period assignment problems, LAGN is the cost of the final Lagrangian function, and FINAL is the total cost of the final feasible assignment.

The optimal solution to the problem lies somewhere between the Lagrangian value and the final solution. This difference is called the optimality GAP. Observe that the average GAP per course is 0.167 and that any course with a GAP less than 10 is considered to be a desirable assignment. *i.e.*, they are in their preferred buildings with all required special features. When we analyzed each course assignment, we discovered that there were only 11 courses that had a GAP greater than 10. (The cost of the room assigned in the solution phase was at least 10 greater than the "acceptable" room.) Note that any course with a GAP of 10 that is three periods long gives an added cost of 30. Therefore, all but 11 classes found desirable assignments.

Problem number 4, Tuesday morning, had the largest gap of 104. There were three courses with a solution cost more than 10 greater than their acceptable cost. Figure 2 gives an example of two courses that were extracted from problem 4 and illustrates the type of situation for which the Lagrangian fails. Neither of these two classes had any other feasible rooms. The assignment solution cost for period 2 is 25, since these classes cannot both have room 56. The total cost from the "SOLUTION" is 50 since

any feasible solution must place one of the classes in a room with a cost of 25 in both periods. This produced a GAP of 25 between the Lagrangian and the final solution. The Lagrangian correctly identified room 56 as an "acceptable" alternate for both classes, but the solution with a GAP of 25 is optimal.

Table 1. Solution Cost

Problem	CLASSES	ASSIGN	LAGN	FINAL	GAP
1	287	989	1020	1072	52
2	220	349	368	408	40
3	13	0	0	0	0
4	188	487	529	633	104
5	164	502	511	530	19
6	21	29	29	29	0
7	283	2105	2153	2173	20
8	240	625	679	730	51
9	17	0	0	0	0
10	181	843	871	882	11
11	170	514	555	621	66
12	11	5	15	15	0
13	253	988	1005	1007	2
14	144	378	379	379	0
Total	2192	7814	8114	8479	365

Rooms:

		13	49	56
Class	Periods			
26	1,2	25	X	0
78	2,3	X	25	0

Figure 2. An Undetected Infeasibility in Problem 4

We considered the idea of adding more complex constraints to the Lagrangian function in order to eliminate this type of assignment. Specifically, in the given example, room 13 is considered as an alternate solution for class 26 in period 2, but it is not an alternate in period 1. (i.e., the period 2 solution must place class 26 in room 13 or class 78 in room 49). Therefore, we could add a constraint that penalizes the cost of assignments that are not valid alternates in all periods for any given class. In this way, all valid alternate assignments would eventually coincide in all periods.

We rejected this option since it would involve a great many additional Lagrangian terms and dramatically increase the required number of iterations. Moreover, since our solutions already appeared to be very close to optimal, we felt that the extra computation was not justified. We conjecture that these constraints would improve the Lagrangian bound and allow us to reduce the solution gap, but they wouldn't have much impact on the final answer. In problem 4, we accounted for an added cost of 63 by locating similar examples to the one described above.

5.2 The Validity of the Cost Model

Statistics were collected on both the manual and the computer assignments and a variety of comparative reports were produced. These reports, which are described below, formed the basis of a series of presentations to the academic departments and lead to the acceptance and implementation of the system in May, 1987 for the Fall (Sept.) term, 1987.

When evaluating the results of any Management Science project, the analyst must avoid the temptation of comparing the total cost of the manual system with the total cost of the algorithm solution. The computation of cost is a part of the model: it represents an approximate measure of the quality of an assignment. The fact that the computer solution produced a lower cost answer than the manual assignment does not necessarily imply that the computer solution is better. Therefore, our comparisons will address the individual components which more accurately reflect people's preferences.

The assignments made by both the manual and the computer systems are referred to as "Initial Room Assignments". Both systems attempt to find the best possible room. After the rooms have been assigned, the Scheduling office begins a "Negotiation Phase" with the department representatives. Some people will change the times of their classes in order to obtain a better room assignment. Others may make arrangements to swap their assignments. However, the majority of assignments remain the same.

Table 2. Building Group Assignments

Group Number	0	1	2	3	inf
Manual Assignment	1111 (77%)	172 (12%)	72 (5%)	12 (1%)	68 (5%)
Computer Assignment	1222 (84.5%)	159 (11%)	42 (3%)	6 (.5%)	17 (1%)

The first criterion measured the preferred building groups. Table 2 compares the assignment of the four groups of buildings for each course in order of preference. The last column labelled 'inf' counts the number of infeasible assignments. In the manual procedure, the scheduling office used buildings for some classes which were not listed in the building table. Generally, they ran out of the required type of space in the preferred buildings and made an undesirable assignment. The computer routine is not allowed to make any assignment outside of the table. Those classes listed under 'inf' are classes for which no feasible assignment was possible. These classes must be handled manually after the fact by asking people to change their class time or accept an unlisted building.

There were 57 different subjects offered at Waterloo. Each subject had an average of two buildings specified in their building group 0. Observe that the computer system assigned over 95% of all classes to group 0 or group 1 compared with 89% in the manual system. Hence, the algorithm appears to be very effective in placing classes in their preferred locations. In addition, there were 41 classes which listed a specific preferred building as an exception to the table values. The manual system satisfied 40 such requests while the automatic version met all of them.

The space utilization factor is more difficult to compare directly. The Scheduling office assigned 71 of the 1400 courses into rooms that were smaller than the estimated class size. These rooms were considered unacceptable by the automated routine. This gave the manual system an unfair advantage in terms of the total number of seats that were left empty. Therefore, we removed these 71 courses from both the manual and the computer results and compared the amount of extra space for the reduced set of courses.

Instead of counting the number of vacant seats, the space penalty for empty seats was used. In this measure, every class is allowed up to ten extra spaces before any cost is incurred. A cost of one is added for every ten empty seats. In the manual system, the total penalty for empty seats was 2309 representing 23,090 empty seats. Since each class is allowed an average of five empty seats without penalty, and there were about

2300 classes compared, this cost implies an average of 15 empty seats per class. The computer system produced a cost of 1665 or 16,650 empty seats, which represents an average of 12 empty seats per class. The computer version left 20% fewer empty seats in classrooms. As noted earlier, this allows for greater flexibility in the manual "room negotiation" phase that follows initial room assignment. Although these results are biased in favour of the computer approach by the elimination of the 71 classes above, we still believe that our method represents a significant improvement.

In the Winter term, there were a total of 364 requests for "essential" special facilities of which the Scheduling Office satisfied 319. The computer system satisfied 351. For the unsatisfied requests, the manual system substituted an apparently unacceptable room while the computer system did not assign any room. Although the computer approach appears to be making better assignments, there is some question concerning the nature of the requests. In several cases, the Scheduling Office *knew* that some requests were not really essential. Therefore, they used some judgement in selecting an appropriate room. This highlights the importance of a manual review of the automated assignments.

A total of 281 courses specified a particular "preferred room". The manual system satisfied 66 of these requests while the computer method assigned 137. The dramatic difference between these results implies that the algorithm is putting a greater emphasis on this factor than the manual system did. In an earlier version of the cost model, a penalty of 1 was added to all non-preferred rooms. In that test, the algorithm only assigned 21 preferred rooms. Perhaps the current penalty of 5 is too high. Recall that this is only a room "preference": if an instructor must have a specific room, it can be "predetermined". Increasing the value of preferred rooms must have a corresponding negative effect on the other cost factors. However, since the other criteria all appear to be reasonable, it was decided to continue using a penalty of 5.

There was some concern that only half of the requests for preferred rooms were satisfied. In analyzing those classes that did not get their preferred assignment, we discovered that many people were asking for the same rooms. Some of these classrooms had already been predetermined by another course. It is not surprising that a few rooms are particularly popular. For the remaining cases, there were two or more requests for the same room and the "winner" was the class with the highest alternate room cost. The tradeoffs, at present, appear to be reasonable.

In the manual assignment, there were 48 examples of room fragmentation where a course did not receive the same room in all meets. One of these involved double fragmentation where a course that met three times a week used three different rooms. The computer assignment produced 58 instances of fragmentation of which one was double. Of the 58 cases, 14 were "intentional" in that the Monday assignment was not particularly good, and the Wednesday/Friday assignment was in a much better room (one or two buildings closer).

A few cases of fragmentation occurred when the assigned room on Tuesday, say (for a Tues./Thurs. course), was predetermined by another course on the Thursday. These situations could have been avoided by initially penalizing any rooms that were not available on all required days. This modification has been incorporated in a later version of the routine.

In both the manual and the computer results, just over half of the courses with fragmented rooms involved non-standard timeslots. As mentioned earlier, this is considered unavoidable when the times are very staggered through the week. We felt that there was too much emphasis being placed on fragmentation since it necessarily adversely affects the other criteria. However, for purposes of implementation, it was considered important to produce assignments that were not dramatically different from the existing

methods.

6. CONCLUSIONS

The algorithm has now been used successfully for four terms: In the Fall 1987 term, there were only six classes that were infeasible compared with 17 for the parallel test. The same results occurred in the run for Winter 1988. This improvement is entirely attributable to better data and careful verification of requests by the Scheduling Office.

Perhaps the most notable endorsement of the algorithm is that nobody noticed. After four terms of use, the vast majority of the faculty and students appear to be unaware that classroom assignment at Waterloo has been computerized. In the Fall term, only twelve of the initial assignments were changed. We assume that this indicates that the new assignments are perceived as being at least as good as the old manual system.

From the viewpoint of the Scheduling Office, the new system has reduced the time and effort required substantially. They estimate that a total of 27 weeks work was required annually. The algorithm became much more valuable in the Fall term of 1988. At that time, a new building opened and forty of the existing classrooms on the campus were replaced by thirty new rooms (with an overall increase in the number of seats). The forty that were removed were converted to lab and office space around all departments while the new rooms were concentrated near the new building. This dramatic redefinition of the space profile would have been very difficult to cope with manually.

It was worth mentioning that the old manual system at Waterloo was considerably better than the average North American university. In these days of severe underfunding and overcrowding, the classroom assignment routine could be very useful at other institutions where they are presently having trouble finding any feasible solution.

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